University of Florida, Department of Mathematics

Algebra First-Year Exam, January 2017, Part 1

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Let G be a group.

(i) If $H \subseteq G$ is a normal subgroup and K is a characteristic subgroup of H, show that K is normal in G.

(ii) Give an example of a group and subgroups $K \subseteq H \subseteq G$ with H normal in G, K normal in H but not normal in G.

2. Let G be a group.

(i) Show that if G is abelian, the subset H of elements of finite order is a subgroup.

(ii) Give an example of a (nonabelian) group G and elements $x, y \in G$ such that xy has infinite order.

3. Show that a group of order $3^2 \cdot 17$ is abelian.

4. This problem has three parts.

(i) Define what it means for a group G to be nilpotent.

(ii) Show that any nilpotent group is solvable.

(iii) Give an example of a finite solvable group that is not nilpotent.

5. Let G be a group acting on a set S, and $H \subseteq G$ a normal subgroup. Assume that for any $x_1, x_2 \in S$ there is a unique $h \in H$ such that $h(x_1) = x_2$. For $x \in S$ let G_x be the stabilizer of x. Show that for any $x \in S$, G is a semi-direct product of G_x and H.

6. Show that for any $n \geq 5$, A_n is the only nontrivial proper normal subgroup of S_n.

7. Show that the normalizer of a 5-Sylow subgroup of A_5 has order 10, and is a maximal subgroup of A_5.